

Heart Physiology

1- All cardiac valves are normally closed during which of the following phases of the cardiac cycle?

- ✓ isovolumetric relaxation
- atrial contraction
- systolic ejection
- ventricular filling

2- The T wave of the electrocardiogram begins to occur during

- early diastole
- isovolumetric relaxation
- isovolumetric contraction
- ✓ systolic ejection

3- Ventricular muscle is isolated and set up in an apparatus to study shortening contractions. At a fixed preload and inotropic state, decreasing the afterload on the ventricular muscle fiber

- increases active tension development by the fiber
- ✓ increases the rate of fiber shortening
- decreases the amount of fiber shortening during contraction
- increases Vmax when shortening velocity is plotted against afterload

4- During the process of cardiac excitation-contraction coupling

- calcium enters the cell through ryanodine receptors
- ✓ calcium enters the cell and stimulates calcium release by the sarcoplasmic reticulum
- ATP is required for contraction but not relaxation
- ATPase bound to actin hydrolyzes ATP to supply energy

5- The "C" wave of atrial pressure curve is due to

- atrial diastole
- ✓ ventricular systole
- ventricular diastole
- atrial systole

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6- Calcium is transported out of the cardiac cell by 3. Phase 4 of SA nodal action potentials is generated in part by

the Ca^{++} -ATPase pump directly coupled to the Na^+/K^+ -ATPase pump

passive diffusion down its concentration gradient

the Na^+/K^+ -ATPase pump

✓ the $\text{Na}^+/\text{Ca}^{++}$ -exchanger

7- The region of the heart with the slowest conduction velocity is the

✓ AV node

His bundle

ventricular Purkinje fiber

atrial muscle

8- The mitral valve normally opens at the beginning of diastolic filling because

the papillary muscles contract

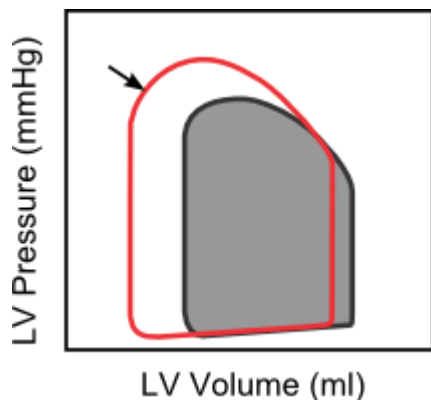
the papillary muscles relax

left atrial contraction forces open the valve

✓ left ventricular pressure falls below the left atrial pressure

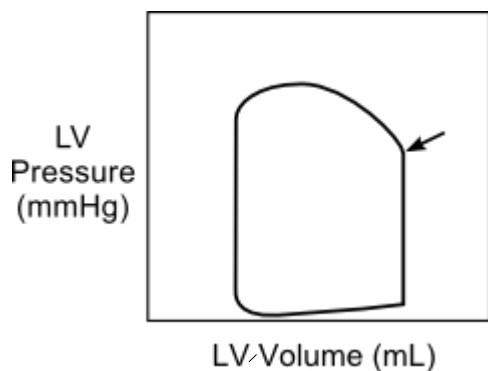
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- 9- Compared to a normal left ventricular pressure-volume loop (grey), the red loop indicated by the arrow represents (Guyton p:118)



- increased stroke volume and decreased ejection fraction
- increased stroke volume with increased end-systolic volume
- ✓ increased inotropy and afterload, and decreased preload
- increased preload, afterload, and inotropy

- 10- The following figure represents a normal left ventricular pressure-volume loop. The arrow is pointed to what event in the cardiac cycle? (Guyton p:118)



- end of systolic ejection
- ✓ aortic valve opening
- mitral valve opening
- end-systolic volume

11- The duration of the effective refractory period in cardiac cells can be increased by

- shortening phase 2
- opening L-type calcium channels
- shortening phase 3
- ✓ inhibiting potassium channels

12- Which of the following leads has an electrical axis of -30 degrees

Lead I

✓ Lead aVL

Lead aVF $\propto 90^\circ$

Lead III

AVR -150

13- The last portion of the heart to depolarize is

- the inner surface of the apex
- the endocardial areas
- ✓ posterobasal portion of the left ventricle and the pulmonary conus
- the entire outer surface of the apex

14- Phase 3 of non-pacemaker action potentials results primarily from

- increased calcium influx through L-type channels
- decreased activity of the Na⁺/K⁺-ATPase pump
- ✓ increased potassium conductance
- inactivation of fast sodium channels

15- In a Lead I ECG recording, the first small negative deflection of the QRS (the Q-wave) represents

- atrial repolarization
- a wave of depolarization traveling towards the left leg
- ventricular repolarization
- ✓ depolarization of the interventricular septum from left to right

16- Closure of the aortic and pulmonic valves produces which heart sound?S₃S₂S₄S₁**17- Ventricular systole is defined as the time between the**

closing of the mitral valve and the closing of the aortic valve

closing of the aortic valve and the opening of the mitral valve

opening and closing of the aortic valve

opening of the aortic valve and the opening of the mitral valve

18- Ventricular preload is increased in response to

decreased blood volume

increased venous compliance

decreased atrial contractility



decreased ventricular inotropy

19- Phase 4 of SA nodal action potentials is generated in part by

opening of potassium channels



funny" currents produced by inward movement of positively charged ions

increased activity of the Na⁺/K⁺-ATPase pump

increased outward movement of calcium ions

20- Conduction velocity in non-nodal cardiac cells is decreased by

sympathetic nerve activation

activation of fast sodium channels

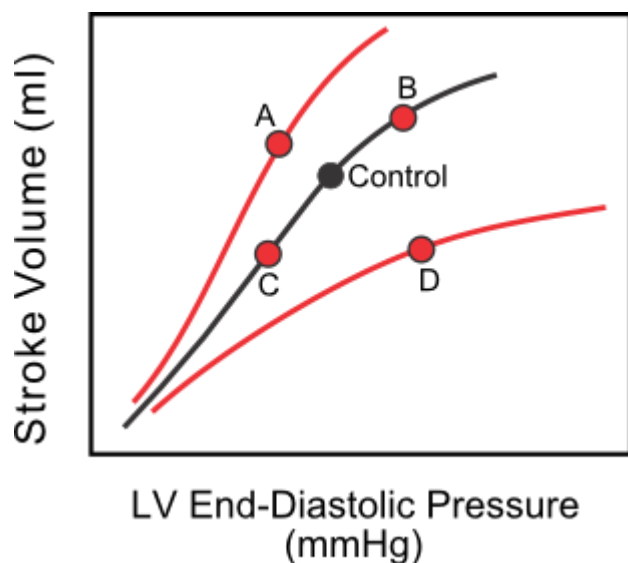
activation of SERCA 2a



decreasing the slope of phase 0

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21- In the figure, left ventricular stroke volume is plotted against left ventricular end-diastolic volume. Relative to the normal control point, which one of the following points represents the independent effects of a decrease in blood volume? (Guyton p:118)



point B

point A



point C

point D

22- The T wave of the electrocardiogram is an electrical representation of

atrial repolarization



ventricular repolarization

atrial depolarization

ventricular depolarization

23- A hospitalized patient has an ejection fraction of 0.4, a heart rate of 100 beats/min, and a cardiac output of 3.5 l/min, what is the patient's end-diastolic volume?

37.5 ml



87.5 ml

92 ml

14.5 ml

24- Leads V₅-V₆ overlie which region of the left ventricle

- ✓ anterolateral
- anteroseptal
- inferior
- anteroapical

25- The initial depolarization during a cardiac muscle action potential results from

- an increase in the relative conductance of potassium
- ✓ a large increase in intracellular sodium concentration
- opening of fast sodium channels
- a large decrease in intracellular potassium concentration

Circulatory Physiology

26- During use *stethoscope*, diastole is heard after phase ... which changing the sound mode is heard.

- 1
- 2
- ✓ 4
- 3

27- Which vessel has most capacity?

- Arteriole
- Capillary
- ✓ Veins
- Artery

28- In circulatory system, there are connections among:

- Liver, heart, blood
- ✓ Vessel, heart, blood
- Skin, heart, blood
- Liver, vessel, blood

29- Which vessel has most capacitance?

- ✓ Veins
- Artery
- Capillary
- Arteriole

30- Normal static pressure in our vessels is around:

- 2-20 mmHg
- 2-200 mmHg
- 1-10 mmHg
- ✓ 7-8 mmHg

31- If a patient's systolic pressure is 100 mmHg and diastolic pressure is 40 mmHg, calculate mean arterial pressure (MAP).

- ✓ 60
- 22
- 70
- 20

32- Electro magnetic flow meter is based on:

- Laplace's law
- Starling's law
- ✓ Fleming's right-hand rule
- Poiseuille's law

33- Which one about chemoreceptors are correct:

- They are distributed in capillary mostly.
- They are only sensitive to oxygen changes.
- ✓ Send signals to vasomotor center of the brain stem
- Send signals to cortex only.

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34- If a patient's systolic pressure is 100 mmHg and diastolic pressure is 60 mmHg, the pulse pressure is

50

100



40

20

35- The great effector in nervous system control of coronary blood flow is....

Parasympathetic system

Oxygen

Adenosine



Sympathetic system

36- Based on Poiseuille's equation, increasing radius of vessels ...

Has no role on flow of vessel

Decrease flow of vessel



Increase flow of vessel

Increase length of vessel

37- Which factor could decrease blood viscosity?



Anemia

Polycythemia

Erythropoietin secretion

Smoking

38- Highest pressure during isovolumic relaxation of heart is ... mmHg.

5

80



120

20

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39- Based on Reynolds law, if a vessels Reynolds number is more than 2000, the blood flow will be ...

Reynold



Turbulent

Laminar

Intermittent

40- In humoral control of circulatory system, which one is considered as vasodilator?

Epinephrine



Histamine

Endothelin

Neurokinin Y

41- Net filtration pressure in capillary bed is around:

33

7

20



13

42- In humoral control of circulatory system, which one is considered as vasoconstrictor?



Anti-diuretic hormone

cGRP

Histamine

Nitric oxide

ADH

43- Conductivity of a vessel could be increase during:

Enhancement of resistance



Enhancement of radius of vessel

Increasing blood viscosity

Increasing length of vessel

44- Which one cannot cause edema:

Reduction in plasma proteins



decrease in capillary permeability

Venus return disorder

Increase in capillary pressure

45- Highest pressure during isovolumic contraction of heart is ... mmHg.

120

20

5



80

46- Baroreceptor reflex are mostly sensitive to



Oxygen changes (60-140 mmHg)

CO₂ changes (40-60 mmHg)

CO₂ changes (1-20 mmHg)

Oxygen changes (10-40 mmHg)

47- About coronary blood flow, which sentence is correct.

Normal coronary blood flow is about 10% of cardiac output.

Coronary blood flow is highest during systole.

Coronary blood flow is lowest during diastole.



Phasic changes are happened in coronary blood flow during systole and diastole.

Blood Physiology

48- What are two elements necessary for maturation of RBCs?



Folic acid and Vit B12

Vit B12 and Vit B

Succinyl-CoA and Iron

Succinyl-CoA and Vit C

49- What happens in the forth line of body defense against infection?

Tissue macrophage invasion

Neutrophil invasion of the inflamed area

☒ Increased production of granulocytes and monocytes

"Walling-off" effect

50- Which type of WBCs is found in blood in the highest number?

Plasma cells

Lymphocytes

Basophils

☒ Neutrophils**51- A patient is suffering from hypoxia. When does the maximum production of erythropoietin occur?**

Within minutes to hours

Within 2 days

☒ Within 24 hours

One week later

52- Which of the following anemia is hemolytic one?

Aplastic anemia

☒ Spherocytosis

Megaloblastic anemia

Blood loss anemia

53- Where is fibrin stabilizind factor secreted from?

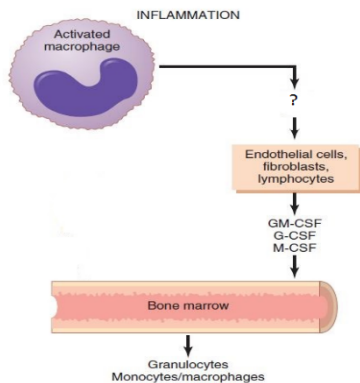
Injured vascular wall and damaged tissues

WBCs

RBCs

☒ Plateletes

54- In control of bone marrow production of granulocytes and monocyte-macrophages, what should be written instead of question mark?



Selectin

IL-6

✓ TNF- α

ICAM-1

55- Immediate effect of a rupture in a blood vessel wall is

Intrinsic pathway of blood coagulation activation

✓ Vascular constriction

Platelet plug formation

Extrinsic pathway of blood coagulation activation

56- The stronger chemotaxis is exhibited by

T Lymphocytes

✓ Neutrophils

Basophils

Eosinophils

57- In thrombocyte adhesion, which factor is maily involved?

ADP

Thrombosthenin

Thromboxane A2

✓ Von Willebrand factor

58- Which type of WBCs releases heparine into the blood?

Monocytes



Basophils

Macrophages

Eosinophils

no WBC

59- The first effect of leukemia on the body is

severe anemia



infectious diseases

a bleeding tendency caused by thrombocytopenia

excessive use of metabolic substrates by the growing cancerous cells

Anatomy

60- The body of 11th vertebra is articulated with all the following bones except ..

Body of 10th vertebrae

head of right and left 11 ribs

Body of 11th vertebrae



Tubercle of right and left 11 ribs

61- Where to insert the text or body of the question

9th

4th



7th

5th

62- All the following veins are drained into the azygous vein except .



The first left intercostal vein

Hemigygous vein

Right subcostal vein

Accessory Hemigygous vein

63- The diaphragm is connected to all the following parts except ...?

The body of the right second lumbar vertebra

The body of the left second lumbar vertebra

Zyphoid process

✓ Transverse process of 10 thoracic vertebrae

64- All the following arteries originate from the internal thoracic except ..

The first anterior intercostal artery

✓ Subcostal

Musculo phrenic

The third anterior intercostal artery

65- In radiographic images, the shadow of the lower side of the heart is related to all the following parts, except ..

The apex of the heart

The left ventricle

The right ventricle

✓ The right auricle

66- The body of the ninth thoracic vertebra is articulated with the head of which of the following ribs

8th and 9th ribs

9th and 10th ribs

8th rib

✓ 9th rib

10
11
12
Just with that
no.

67- Which of the cardiac cavity is anterior compare to the others in radiographic images?

The left atrium

The left ventricle

✓ The right ventricle

The right atrium

68- Lymphatic of all the following areas drainage into the thoracic duct except ..?

The right lower limb

The ninth right intercostal space

Left lower limb

✓ The second right intercostal space

69- All of the following muscles are innervated by the posterior (dorsal) branch of the intercostal nerves except ..

✓ serratus posterior inferior

Longissimus

Multifidus

Ilio Costalis thoracis

70- The phrenic nerve is involved in innervating all the following structures except ..?

Parietal serous pericardium

✓ Visceral serous pericardium

Fibrose pericardium

Diaphragm

71- Which of the following ligaments connects to the spinous process?

✓ Ligamentum nuchea

Ligamentum flavum

Anterior longitudinal ligament

Posterior longitudinal ligament

72- In the sympathetic chain, all of the following definitions are correct except?

The sympathetic chain is also observed in the abdomen

The white communicating branch is on the lateral side of the gray communicating branch

Large splanchnic nerves contain myelin

✓ The white communicating branch is visible along the entire length of the sympathetic chain

73- Which of the following structures is more to the left of the body than the rest?

Left recurrent laryngeal nerve

Left ventricle

Left vagus nerve

✓ Left phrenic nerve

74- All the following muscles are connected to the fourth rib except ...

Pectoralis major

✓ Latissimus dorsi

Serratus posterior superior

Pectoralis minor

75-

Which of the following elements is adjacent to the anterior wall of the esophagus?

Hemi azygous vein

Thoracic duct

Right vagus nerve

✓ Pericardium

76- In the right dominate heart which artery is located in the anterior ventricular sulcus?

✓ A branch of the left coronary artery

A branch of circumflex A

Right marginal artery

A branch of the right coronary artery

77-

In which cardiac cavity is the septum marginal (moderator band) seen?

Left atrium



The right ventricle

Right atrium

The left ventricle

78- In which of the following joints is there an intra-articular ligament?

Zypho sternal joint

First Costo vertebral joint



The second costo-sternal joint

Costo transvers joint

79- The opening of all the following veins open to the right atrium except ..



Pulmonary veins

Superior vena cava

Inferior vena cava

Coronary sinus

80- The ascending aorta is adjacent to which of the following structures?

Vena cava superior

The left pulmonary vein

The right atrium



Infundibulum

81- Which of the following structures is adjacent to the aortic arch at the bottom?

Left vagus nerve

Right vagus nerve



Superficial heart network

left phrenic nerve

82-

All the following structures are seen in the posterior wall of the heart except ...?

Coronary sinus



Conal artery branch

Middle cardiac vein

Circumflex artery

83- All of the following structures are seen in both the upper mediastinum and the posterior mediastinum except ...?

Right vagus



Left recurrent laryngeal nerve

Esophagus

Left vagus

Histology

84- Which antibody has pentamer structure?

Ig E

Ig A



Ig M

Ig G



85- Which one is the antibody of secretions such as saliva?



Ig A

Ig E

Ig G

Ig M

86- Which organ has dual embryonic origin?

Spleen

Lymphnode

Lingual tonsil

✓ Thymus

87- Where is the site of blood filtration in body?

✓ spleen

tonsil

thymus

lymphnode

88- Most numerous vessels in our body are

Small arteries

✓ capillaries

Large arteries

veins

89- The thickest layer in the wall of veins are....

✓ adventitia

media

subendothelia

intima

Exa

90- Which one is the largest lymphatic organ in the body?

Lymphnod

Palatin tonsil

✓ Spleen

thymus

91- Where we observe Hassall's corpuscle in thymus?

Cortex of lobule

capsule

terabecula

✓ Meulla of lobula

92- Which one is primary lymphoid organ?

✓ Thymus

Lymphnode

Spleen

Tonsil

93- In which vessel we observe internal elastic layer obviously?

Elastic artery

Medium sized veins

✓ Muscular artery

capillary

Embryology

94- Failure of the aorticopulmonary septum to undergo the 180-degree spiral will result in

An atrial-septal defect

Tetralogy of Fallot

✓ Transposition of the great vessels

.Persistent truncus arteriosus



f

95- The right ventricle of the heart is formed primarily from which embryonic structure?

Primitive ventricle

Sinus venosus

Primitive atrium

✓ Bulbus cordis

96- The right subclavian artery is formed from all of the following vessels EXCEPT:

7th Segmental arter



Right 6th aortic arch

Right dorsal aorta

Right 4th aortic arch

97- Which statement is TRUE about interatrial septum development?

The foramen (ostium) secundum is an opening in the septum secundum.

The foramen secundum is also known as the foramen ovale.

The septum primum is thick and muscular, compared to the thin and membranous septum secundum.



The foramen (ostium) primum disappears when the septum primum and inferior endocardial cushions fuse.

98- Which statement is CORRECT about heart tube folding?

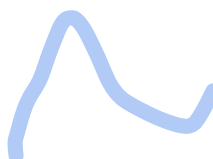
The sinus venosus moves superiorly and anteriorly



The primitive atrium moves superiorly and posteriorly

The bulbus cordis moves inferiorly, posteriorly, and to the embryo's right side

The primitive ventricle moves to the embryo's right side



99- In the fetus, blood bypasses the liver by traveling through which vessel?

ductus arteriosus

umbilical artery

hepatic portal vein



ductus venosus

100- Which part of the primitive heart tube gives rise to the pulmonary artery and the aorta?

Primitive ventricle



Truncus arteriosus

Bulbus cordis

Sinus venosus